

# Nameer Iqbal Ansari

RTL Engineer | ASIC | FPGA | RISC-V

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## PROFESSIONAL SUMMARY

RTL Design Engineer (FPGA/ASIC) at XCLR Technologies, specializing in **RISC-V-based SoC** development. Certified **RISC-V Foundational Associate (RVFA)** and **Google Summer of Code (GSoc) 2022 contributor** and **2023 mentor**, with hands-on experience in **SystemVerilog/Verilog** based **ASIC** and **FPGA** designs, **Lint/CDC/RDC** sign-off, **timing closure**, and **verification**.

## WORK EXPERIENCE

02/2025 – Present

**RTL Design Engineer, XCLR Technologies (Pvt.) Ltd.**

- **High-Speed I/O Subsystem (HSIOSS)** — Led design and verification across **3 teams** and **6 engineers** (excluding IP vendor teams) for a subsystem integrating a **32 Gbps SerDes** and **PCIe Gen5** controller, running at a **1.5 GHz** core clock with a **100 MHz** reference clock and **2048-bit data / 64-bit** configuration **AXI interfaces**. PIPE interface between **MAC** and **PHY** spanning **8 PCIe lanes**, each interfacing through a **32-bit** PIPE datapath. Defined the verification test plan including directed chip-level firmware testing alongside block-level SystemVerilog testbenches. Also utilized and modified the existing UVM testbenches within the IPs
- **RISC-V Vector Load/Store Unit (VLSU)** — Designed a **2048-bit** RVV load/store unit interfacing to a **64-bit** memory port, with **AXI-ID tracking**, address generation and translation, and data packing/unpacking logic supporting configurable element widths (**8–64 bits**) within a single vector register. Added multi-stage **pipelining** to **reduce latency** and **cell count**
- **IP Replacement Across 5 Subsystems** — Re-architected 5 subsystems by replacing Synopsys-sourced data/protocol converters, crossbars, **CDC** synchronizers, and **AXI-to-native-memory bridges** with PULP Platform IP. Utilized Synopsys **CoreAssembler** to generate a GUI-based integrated RTL, and also used **CoreConsultant** to generate standalone Synopsys IPs
- **Control-Register-to-AXI Bridge** — Designed and verified a protocol bridge between a **16-bit** native SerDes control-register interface and a **64-bit** AXI configuration bus, resolving **CDC** and 64-to-16-bit **data-width conversion**. Validated the design with **block-level SV TB** and **system-level firmware** testing after integration of the converter into the chip
- **512-bit eFuse Controller** — Implemented **32-bit** AXI access by bridging the native **eFuse** memory array interface to an **AXI-based** register-access architecture, including command-driven read/write transaction handling. Performed SystemVerilog-based block-level testing and validated end-to-end chip-level functionality through **QSPIS** interface access testing
- **Chip-level low-power implementation** — Performed PPA analysis and enabled up to **80%** of the chip to operate in low-power mode using **low-power** features provided by Synopsys IPs and custom design logic. Created a **safe power-switch-off** mechanism that ensured **glitch-free clock gating** for low-power entry and in-flight **AXI** transaction handling by safely handling incomplete or unintended AXI transactions via AXI response error (**SLVERR/DECERR**) functionality during power-state transitions
- **VU19P FPGA SoC Bring-Up** — Led development and emulation of an SoC targeting the VU19P FPGA using **Xilinx Vivado** for **synthesis**, implementation, and **bitstream** generation. Integrated UART, DDR, and JTAG interfaces, and validated DDR and JTAG functionality through on-hardware emulation testing on the VU19P board
- **PULP BSCANE2 TAP Integration for Unified JTAG Access** — Integrated PULP's **dmi\_bscane\_tap** via **BSCANE2**, enabling a single **HS-3 cable** on the VU19P's JTAG TAP to handle both bitstream configuration and RISC-V debug access. Validated the path on-hardware using **OpenOCD/GDB**, testing **halt/resume**, **single-stepping**, **breakpoints**, and **register/memory access**
- **DDR4 Bring-Up & Linux Boot on AWS EC2 F1 FPGA** — Interfaced the SoC with AWS EC2 F1's **DDR4-C** shell controller to bring up external memory, then loaded the Linux boot image into DDR and validated end-to-end boot on the Ariane (**CVA6**) RISC-V core

09/2023 – 02/2025

**Associate Engineer, Xcelerium**

- **12nm timing closure at 1.5 GHz of RISC-V-based core subsystem** — Achieved **-0.01** worst negative slack (**WNS**) and reduced setup and hold (**reg2out/in2reg/in2out**) violations across **500+** violating paths by optimizing combinational logic, skew balancing, registering, floorplanning, and SDC constraint fixes
- **JTAG daisy Chain at Aries** — Designed and validated a **daisy-chained JTAG** debug architecture linking two PULP **riscv-dbg TAP** controllers across two subsystems, allowing either core to be halted and debugged independently while the other ran normally. Verified via **SimJTAG (C DPI)** in simulation and on **AWS EC2 cloud FPGA**
- **Lint, CDC, and RDC sign-off** — Accomplished **0 error/warning** sign-off across 4 subsystems, each containing a minimum of **100** violations that are detected using **Synopsys SpyGlass**, ensuring digital designs are ready for physical design (PD)
- **AWS EC2 F1 Shell Interface Validation** — Verified 4 AWS EC2 F1 shell interfaces (**OCL, BAR1, DDR, SDA**) using **C++ drivers** and SystemVerilog testbenches, across both AWS EC2 F1 emulation and simulation environments.
- **Logic-Analyzer-Based Debugging** — Debugged JTAG and SoC subsystem behavior using logic analyzers across both AWS Cloud emulation and VU19P FPGA hardware environments
- **Technical Documentation & Design Reviews** — Authored comprehensive design documents, verification test plans, timing diagrams, and block diagrams to aid design reviews and knowledge transfer

03/2020 – 09/2023

## Research Assistant, *Micro Electronics Research Lab (MERL)*

- **RV32I Core Design** — Built an open-source [RV32I CPU core](#) from scratch in SystemVerilog, simulated with **Verilator** and **FuseSoC**, and emulated on an **Arty A7 FPGA** with **Xilinx Vivado**. Later integrated basic **UVM** and **JasperGold** support
- **Out-of-Order Upgrade to RV32I Core** — Extending the **RV32I** core with out-of-order execution capability, adding 2-wide instruction fetch, register renaming via a **Register Alias Table (RAT)**, a reorder buffer (**ROB**) for in-order commit, and reservation stations feeding ALU and branch execution units. Migrating the existing in-order pipeline to a **dispatch/issue/commit** structure inspired by **Berkeley's BOOM** design
- **NOVA1 SoC Development** — Contributed to **NOVA1 System-on-Chip (SoC)** development and its **AWS EC2-based** emulation, and **Zyper OS** bring-up
- **Multicore Cache Coherency & CHI Research** — Led the research of the **multicore architecture** and its **cache coherency** through the **OpenPiton** framework. Also researched converting OpenPiton's native coherence protocol to ARM's **CHI**, mapping its directory-based coherence engine onto CHI's **RN-F/HN-F/SN-F** node abstractions
- **RISC-V Training & Mentorship** — Conducted **student training programs** on RTL design and how to contribute to open-source

## TECHNICAL SKILLS

- **RTL Design** — RISC-V VLSU | JTAG | SerDes | PCIe | eFuse | UART | CDC | XBAR | Cache | AXI | APB | TileLink
- **EDA Tools** — VCS | Verdi | Xilinx Vivado | GTKWave | CoreAssembler | CoreConsultant | QuestaSim | Verilator | FusionCompiler | PrimeTime
- **Formal Verification** — JasperGold | SVA (SystemVerilog Assertions)
- **Static Analysis** — Synopsys SpyGlass (Lint, CDC, RDC)
- **Languages** — Verilog | SystemVerilog | SystemVerilog-OOP | UVM (basic) | C | C++ | RISC-V Assembly
- **Scripting** — Tcl | Shell | Makefile | CI/CD | Python | YAML (FuseSoC)
- **Version Control** — GitHub | GitLab
- **Operating System** — Linux | Windows

## SOFT SKILLS

**Languages** — English (C1), Urdu (Native)

**Other** — Design review | Technical Documentation | Debugging & Root-Cause Analysis | Design verification coordination | Technical Leadership | Timing-Closure Problem Solving | Research & Mentorship | Volunteer work

## PROJECTS

### Google Summer of Code (GSoC), Contributor (2022) / Mentor (2023)

- Added **FuseSoC** support on the **Azadi-SoC** and **rebased** the **lbex core** to latest upstream commits ([GSoC'22](#))
- Mentored **TileLink** Uncached Heavyweight (**TL-UH**) implementation in **Azadi-SoC** ([GSoC'23](#))

### MARCore & NOVA1, Research Lead / Contributor

- Researched multicore cache-coherent memory systems using OpenPiton workflow and Ariane Core ([MARCore1](#))
- Contributed to NOVA1 SoC that contains RV64GC Core, UART, BRAM, DDR, and JTAG support ([The NOVA Project](#))

## EDUCATION

2019 – 2023

### Bachelor of Engineering in Electrical Engineering, *Usman Institute of Technology*

(CGPA: 3.289) (Specialization: Computer Systems)

- **Major Courses** — Digital Logic Design | Computer Architecture | Microprocessor-Based Systems | Embedded Systems | Operating Systems

### Honors & Awards

- Hasham Foundation Scholarships by Usman Institute of Technology in 2022
- Linux Foundation (LiFT) Scholarship Recipient in 2024

## CERTIFICATES

**RISC-V Foundational Associate (RVFA)** — Date: Aug 2025 | Issuer: RISC-V International | Sponsor: Linux Foundation | [Link](#)

**RISC-V Fundamental (LFD210)** — Date: Aug 2025 | Issuer & Sponsor: Linux Foundation | [Link](#)

## PUBLICATIONS & ARTICLES

- Multicore SoC For SMP Linux ([Thesis](#))
- NOVA Emulation Project- An open-source RISC-V-based application-class AI SoC ([Project Doc](#))
- A Summer of Opportunity ([Medium — GSoC experience](#))
- Abstractions Can Lead Humanity to Ignorance (An Unpopular Opinion) ([Medium — technical opinion](#))